

```
In [1]: from keras.preprocessing.image import ImageDataGenerator, array_to_img, img_to_array

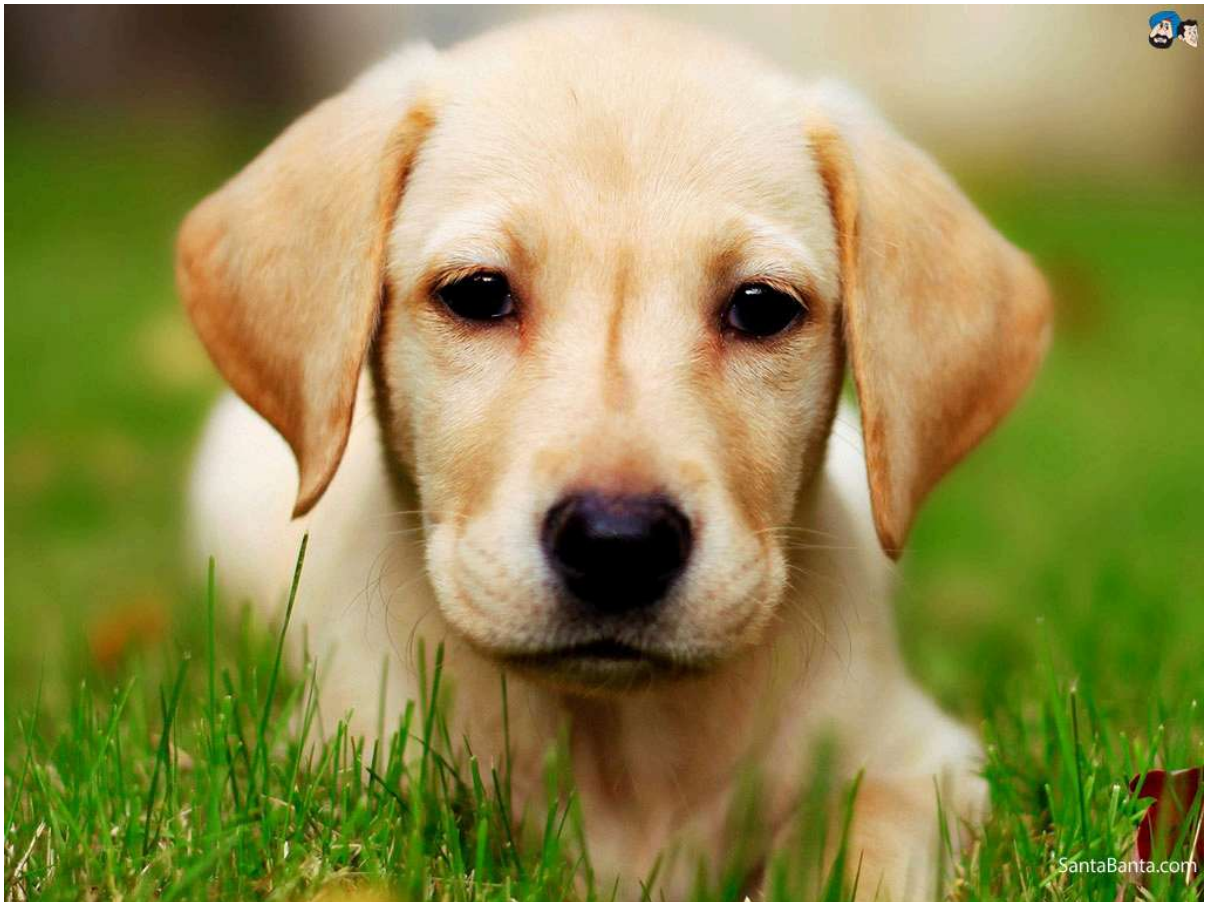
datagen = ImageDataGenerator(
    rotation_range=40,
    width_shift_range=0.2,
    height_shift_range=0.2,
    shear_range=0.2,
    zoom_range=0.2,
    horizontal_flip=True,
    fill_mode='nearest')
```

Using TensorFlow backend.

```
In [2]: img = load_img(r"C:\Users\kdata\Desktop\Datascience Project\Data Augumention\dogs-8
```

```
In [3]: img
```

Out[3]:



```
In [5]: x = img_to_array(img) # this is a Numpy array with shape (3, 150, 150)
x = x.reshape((1,) + x.shape) # this is a Numpy array with shape (1, 3, 150, 150)

# the .flow() command below generates batches of randomly transformed images
# and saves the results to the `preview/` directory
i = 0
for batch in datagen.flow(x, batch_size=1,
                           save_to_dir=r"C:\Users\kdata\Desktop\Datascience Project\
                               i += 1
```

```
if i > 30:  
    break # otherwise the generator would loop indefinitely
```

In []: